MSG No. 127.390 $P4'/mb'm$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,5,6,9,10,13,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,4,7,8,11,12,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'/m..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,5,6,9,10,13,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,7,8,11,12,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `4'..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,9,10]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,4,11,12]`
3	$[0,\ 0,\ -z]$	`[5,6,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[7,8,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,7,8]`
2	$\left[0,\ \frac{1}{2},\ -z\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2},\ 0,\ z\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[11,12,13,14]`

Table 7: Wyckoff site: `4g`, site symmetry: `m.2m`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	`[1,3,6,8]`
2	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	`[2,4,5,7]`
3	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	`[9,12,14,15]`
4	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	`[10,11,13,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `m.2m`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[1,3,6,8]`
2	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[2,4,5,7]`
3	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	`[9,12,14,15]`
4	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[10,11,13,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `m..`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,6]`
2	$[-x,\, -y,\, 0]$	`[2,5]`
3	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	`[3,8]`
4	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, 0\right]$	`[4,7]`
5	$[-y,\, x,\, 0]$	`[9,14]`
6	$[y,\, -x,\, 0]$	`[10,13]`
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	`[11,16]`
8	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, 0\right]$	`[12,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	`[1,6]`
2	$\left[-x,\, -y,\, \frac{1}{2}\right]$	`[2,5]`
3	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[3,8]`
4	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[4,7]`
5	$\left[-y,\, x,\, \frac{1}{2}\right]$	`[9,14]`
6	$\left[y,\, -x,\, \frac{1}{2}\right]$	`[10,13]`
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[11,16]`
8	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[12,15]`

Table 11: Wyckoff site: `8k`, site symmetry: `..m`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, z\right]$	`[1,8]`
2	$\left[-x,\, \frac{1}{2}-x,\, z\right]$	`[2,7]`

continued ...

Table 11

No.	position	mapping
3	$\left[x,\, x+\frac{1}{2},\, -z\right]$	[3,6]
4	$\left[-x,\, \frac{1}{2}-x,\, -z\right]$	[4,5]
5	$\left[\frac{1}{2}-x,\, x,\, z\right]$	[9,15]
6	$\left[x+\frac{1}{2},\, -x,\, z\right]$	[10,16]
7	$\left[x+\frac{1}{2},\, -x,\, -z\right]$	[11,13]
8	$\left[\frac{1}{2}-x,\, x,\, -z\right]$	[12,14]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[3]
4	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$[x,\, y,\, -z]$	[6]
7	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[7]
8	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[8]
9	$[-y,\, x,\, z]$	[9]
10	$[y,\, -x,\, z]$	[10]
11	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[12]
13	$[y,\, -x,\, -z]$	[13]
14	$[-y,\, x,\, -z]$	[14]
15	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[15]
16	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[16]